

SOME RECENT MOTOR DRIVERS

Part Number	Manufacturer	Features
A5989	Allegro Microsystems	Rated up to 1.6A, high-current DC driver at 3.2A, fixed off-time pulse width modulation, current regulators, 2-bit non-linear digital-to-analogue converters
TB67H400A	Toshiba	Two-channel DC motor driver, 50V maximum rating, built-in error detection circuits including thermal shutdown, over-current shutdown, under-voltage lockout
STSPIN220	STMicroelectronics	Two on-chip MOSFET bridges for driving two brushed DC motors, position resolution of 256 micro-steps per step, built-in protection features include over-current, over-temperature, short-circuit protection
DRV8305-Q1	Texas Instruments	Smart gate-drive architecture with programmable slew-rate control, integrated 3.3V or 5V linear regulator, on-chip protection with fault diagnostics guard
IRSM836-084MA	Infineon Technologies	Optimised dV/dt for loss, motor power range up to ~150W, without heat-sink, integrated gate drivers, protection shutdown pin
TLE9867QXA20XUMA1	Infineon Technologies	ARM Cortex M3 MCU, system clock up to 40MHz, NFET drivers with charge pump, on-chip oscillator and PLL
LV8907UW	ON Semiconductor	Sensor-less three-phase motor controller, operating temperature range up to 175°C, gate driver for six N-channel MOSFETs, two-stage charge pump
UCC27211A-Q1	Texas Instruments	Operating voltages up to 100V, fast propagation delay at 15ns, fast rise and fall times of 7ns, 4-A peak gate drive
STSPIN230	STMicroelectronics	Three-phase brushless motor driver, operating voltage from 1.8V to 10V, maximum output current 1.3A rms
A4955/56	Allegro Microsystems	Support voltages up to 50V and load currents of up to 20A, analogue output function, internal pulse width modulation current control, VDS sensing

Note: This is a list of some recent drivers noticed in the market at the time of writing the article. The selection is not based on any other criteria

when to commutate the motor.

To implement features like sensor-less control, automatic and integrated-state machine logic to control the motor, “it requires highly smart algorithms to control the system,” says Anayya Vallabh, technical sales engineer, Texas Instruments (India) Pvt Ltd. He adds, “These should follow the theoretical mechanism to spin the motor.” Besides, it should take care of practical faults that might rise during operation.

Vallabh explains, “For example, the motivation behind sensor-less control is applicability with end usage.” Naveen Bevara, analogue applications engineer, Texas Instruments (India) Pvt Ltd, adds, “Motors typically include what is called a position sensor, which can identify the position of the rotor (moving part of a motor). Identifying rotor position is essential for control of the motor at various speeds and under changing loads.”

Reliability in speed control

One of the innovative uses of motors in applications today (such as air-conditioners, refrigerator compressors, washing machines and cooling

Trends at a glance

- Sensor-less control in BLDC motors
- Automatic tuning system, which seeks optimum parameters of the motor
- Improved power efficiency
- Integration of low power and state machine control

fans) is to drive and control a motor at variable speeds and even turn it off at times to improve product efficiency. The motor must be able to then re-start quickly and reliably 100 per cent of the time. This has allowed manufacturers to improve reliability of their products, improve efficiency by using variable-speed motor technology and eliminate costly position sensors from their systems.

Automatic tuning

Running the driver in stepper motor application requires tuning. If manual tuning is used, it is very difficult to find the accurate point to set while tuning. Auto-tuning is an intelligent system, which seeks optimum parameters of the motor. Auto-tuning suppresses oscillations, making frictional compensation. It does not require any input

The evergreen driver

L293D from Texas Instruments is an absolute favourite among those building robots. Be it a line follower or a war bot—you say motor driver, they say L293D. This 16-pin IC can control a set of two DC motors simultaneously in any direction. However, recently a lot of other drivers are also making their name in this segment.

from you. The motor will perform a sequence of movements in order to determine the type of load that is controlled and the type of coupling. This tuning covers most applications with good performance.

STSPIN motor drivers from STMicroelectronics can be used in a wide range of applications including robotic positioning systems, printer motors, camera autofocus mechanisms, toothbrush motors or syringe pumps other than the intended purpose of portable and wearable design. Sized at 3mm × 3mm, these are being dubbed as the smallest in size.

Power applications

Improved power efficiency is always the guiding factor for technology trends. Bevara explains, “When it